BUILD PLAN

Monte Carlo Underwriting Agent — Rightway Proof Case

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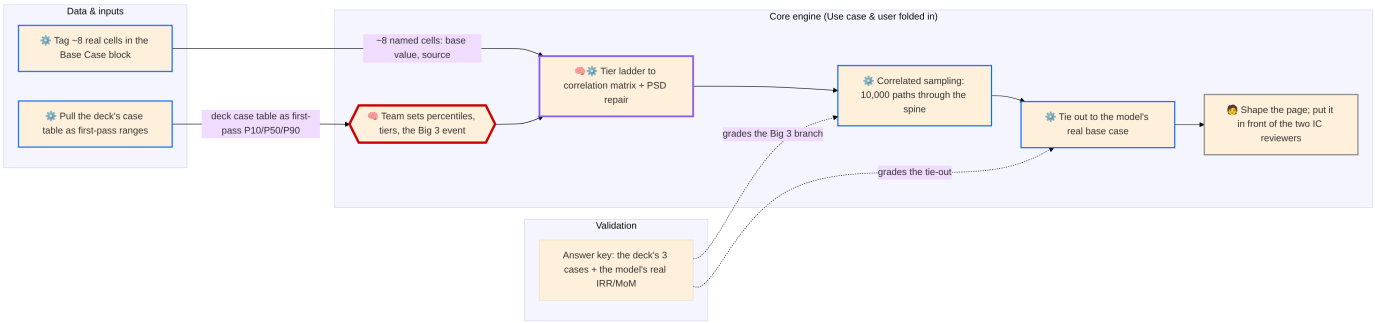
What's Settled

- **Target system:** Claude Skill — a single, repeatable workflow run per deal, no separate backend or database.
- **Prove first:** the tier-ladder-to-correlation-matrix translation plus the correlated simulation engine, tied out against Rightway's own real base case, tested on the one genuinely tricky case — the Big 3 pricing-response event, which has to move a continuous driver *and* correlate with the exit multiple at the same time.
- **Reuse (don't rebuild):** FP's existing DCF Model Builder skill's Excel case-selector convention for the output page; standard numpy / scipy / pandas / openpyxl inside Claude's own code execution — no new tool or license; the Rightway deck's own rule-of-40-to-multiple rationale (slide 28), fit as a curve rather than re-derived from scratch.
- **People & window:** Emilio, Katrina and Lawrence — three deal associates who work in modeling and IC decks day to day; none currently write Python or use Claude Code as builders. Several days together (Aug 4–6), then async to the Aug 11 demo.
- **Already have:** the real Rightway July 2026 Update IC deck and the real Rightway Model v30.xlsx, both in hand on day one. The deck's own base / upside / downside case table is both the example input and the answer key.

One thing worth flagging before the lanes below: opening the actual model changed the wiring plan. Rightway's three cases are not one set of assumption cells flexed by a toggle — they're three separate, fully hardcoded column blocks (Base: columns Q:S, Upside: Z:AB, Downside: AI:AK, in `Opmodel - PBM` and `Opmodel - Nav`), selected by `Summary!AA11` (1/2/3/4). That means tagging happens **inside the Base Case block specifically**, with the case toggle held at 1 throughout every simulated path — not traced through a live flex. This is simpler than the general approach, not harder: the assumption cells inside that block are plain numbers, not formulas.

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1. The Four Lanes



Color key: 🟣 judgment (thick red, the prove-first step) · 🟪 mixed · 🟦 mechanical · ⚪ manual.

- **Data & inputs** ships: the tagged cells and the deck's case table, in an agreed shape.
- **Core engine** ships: the thing that runs — the tier-ladder translation, the correlated sampler, the tie-out, and the output page (Use case & user folded in here).
- **Validation** ships: the score, and where it breaks.

Hand-offs: - Data & inputs → Core engine: the ~8 tagged cells, each with its real base value and a slide or model citation, plus the deck's case table as first-pass P10/P50/P90 by slide. - Core engine (team's own call) → Core engine (the build): the percentiles, linkage tiers, and the Big 3 event definition — the one step that stays entirely the team's judgment, made explicit. - Core engine → the two IC reviewers: one page — the distribution with the deck's three cases marked on it, the probability table, and the best-decile-versus-worst-decile driver read. - Validation ↔ Core engine: the answer key (the model's real base/upside/downside IRR and MoM) grades the tie-out on every change, and grades whether the Big 3 branch shifts the right variables in the right direction.

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2. How We'll Know It Works (and How It'll Break)

What it needs to do

It needs to…	Mark
Turn the team's percentiles and linkage tiers into a valid correlation matrix	EXACT
Sample 10,000 correlated paths, including the Big 3 binary event, through the model's real spine	EXACT
Reproduce the model's own base-case IRR and MoM within 10bps / 0.02x	EXACT
Shift the right variables in the right direction when the Big 3 branch fires	RANGE
Cite a source for every variable's range and every correlation number	EXACT
Produce a page the two IC reviewers find useful without a walkthrough	RANGE

What	"Working" looks like	"Broken" looks like	Where you'd see it	How we check
Core engine — tier ladder + sampling	Matrix passes PSD repair with a small adjustment log; P10/P50/P90 stable over the last 2,000 of 10,000 draws	Repair silently moves a stated linkage by more than 0.10; the band still moving at 10,000	The correlation pack; the convergence chart	Re-run against the fixed tier-ladder table on every change

Core engine — the tie-out	IRR within 10bps and MoM within 0.02x of the model's own base case at every variable's P50	Any spine row diverges from the real model	The tie-out exhibit, row by row	Re-run against the real Rightway model on every change
Core engine — the Big 3 branch	New-deal volume, retention, and (via the exit-multiple link) the multiple all move together when the branch fires, in the direction the deck's own risk table implies	Only one variable moves, or the multiple doesn't follow growth and margin down	The worst-decile input profile on the output page	Validation reads the shifted profile against the deck's risk table
Data & inputs — the tagged cells	~8 named cells, each with its real base value and a slide or model citation	A tag that isn't actually inside the Base Case block, or has no source	The variable register	Spot-check every tag against the workbook before Core builds on it
Use case & user (folded in) — the page	The two IC reviewers read the shape and the drivers without a walkthrough	They ask what a number means, or ask for the old three-case table back	The review session	A real reaction from the two IC reviewers
Whole thing, start to finish	Team's raw percentiles/tiers/event → the page in front of the two IC reviewers, no manual patching	Any step needs a hand-edit to produce the next one	A dry run before the review session	One full run on Rightway, days 1–3; repeated async

3. How the Work Divides

Element of work	Lane	Can start day one on
Tag the ~8 real assumption cells in the Base Case block and confirm each one's citation	Data & inputs	the attached Rightway model
Pull the deck's case table (slide 28) as the first-pass P10/P50/P90 for each variable	Data & inputs	the attached Rightway deck
Set the team's own percentiles, linkage tiers, and the Big 3 event definition	Core engine	the deck's case table as a starting point
Build the tier-ladder-to-correlation-matrix translation, with the PSD repair log	Core engine	the fixed tier-ladder table
Build the correlated sampler and the reduced-form spine, calibrated to the Base Case block	Core engine	the tagged cells, once agreed
Run the tie-out against the model's real base case	Core engine	the attached Rightway model
Shape the output page and book the review with the two IC reviewers	Core engine (Use case & user folded in)	a mocked-up page, before the engine is proven
Write the answer key: the deck's three cases plus the model's real IRR/MoM	Validation	the attached deck and model
Grade the tie-out and the Big 3 branch on every change	Validation	the answer key
Get the two IC reviewers' real reaction and grade it	Validation	the booked review session

Across 3 people: Core engine, with Use case & user folded in, on its own; Data & inputs on its own; Validation

on its own. **Protected split:** Validation never sits with Core engine.

A gap worth naming plainly: none of the three builders write Python or use Claude Code today. That's what the tool is for — Core engine's work happens by directing Claude directly rather than hand-writing the engine, and its hardest day-one skill is reading a tie-out delta and pushing back on what comes back, not syntax. That learning curve is real and deserves more of the in-person time than any single line item below.

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4. Checkpoints — in person, then async

Days 1–3, in person (Aug 4–6) — get the Core working

- **Day 1 (Aug 4):** agree hand-offs first — the ~8 tagged cells' names, base values and citations, plus the deck's case table as first-pass ranges. Core starts on the tier ladder and sampler immediately against the deck's own numbers; Validation starts the answer key straight off the model's real IRR/MoM; Data & inputs tags the cells.
- **Day 2 (Aug 5):** Core against the answer key — tie-out attempted, graded. This is the narrowing day: if the tie-out won't close, cut which cells get tagged now (drop the Nav-tab granularity, hold opex fixed) while everyone is still in the room.
- **Day 3 (Aug 6):** Core holds the tie-out on the base case; prove the Big 3 branch specifically — the one tricky case. Walk every hand-off out loud, and book the one async sync plus the review slot with the two IC reviewers before anyone disperses.

If Core isn't holding the tie-out at the end of day 3: say so plainly — the async half becomes "keep proving Core," not "start wrapping it." Nothing below gets funded until the base case ties out.

Days 4–8, async (Aug 7–10) — wrap the proven Core

Spent in scoring order:

1. **Core still first.** Keep closing the tie-out and the Big 3 branch if either is still wobbling — that's the work, not a nicer page.
2. **Then elegance.** Re-check the correlation tier-boundary robustness test (does the answer move less than ~150bps across a tier's boundary values). Confirm the tier-ladder mapping is doing the team's own judgment, not quietly become a black box that happens to pass the tie-out.
3. **Then completeness, if time genuinely remains.** The audit appendix, a second past deal as a fresh real case, the archetype-driven opening conversation from the fuller workflow spec — all explicitly last.

One scheduled sync: booked before dispersing, placed at the point the two IC reviewers see the real page for the first time — not a status update at the end.

One-line fallback per lane: - Core engine — the tagged cells and the tier ladder are agreed in writing, so the other two lanes keep moving against them even if Core's driver is out for two days. - Data & inputs — the deck's case table already gives Core a first-pass range, so a delay here doesn't block the tie-out attempt. - Validation — the answer key is written from the model directly, so grading can happen without the checker present at every run.

Ship (Aug 11): demo recorded; the output page and the config file — the audit trail — delivered alongside it.